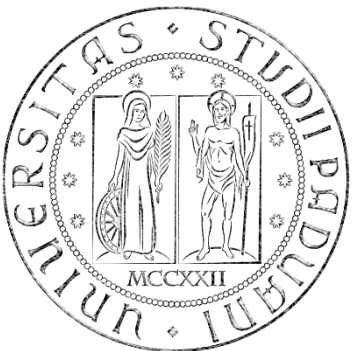


Biomedical Wearable Technologies for Healthcare and Wellbeing

Introduction

A.Y. 2025-2026
Giacomo Cappon



Who am I

➤ Giacomo Cappon

- Assistant Professor, Department of Information Engineering, University of Padova
- PhD in Information Engineering, curriculum Bioengineering
- Main research interests: signal processing; decision support systems; wearable sensors; digital health; ai for medicine



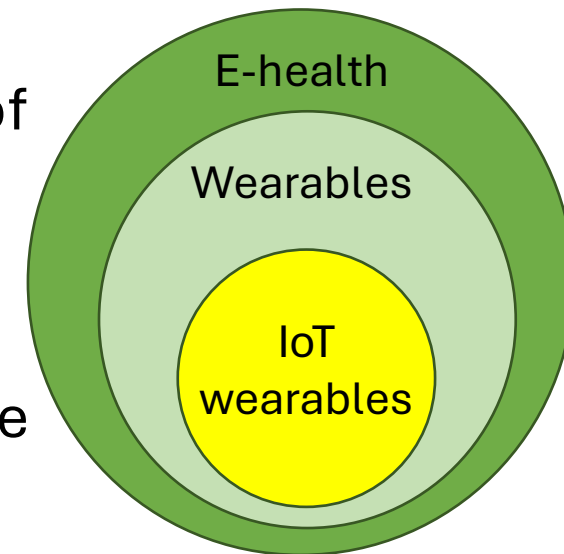
➤ Contacts

- email: giacomo.cappon@unipd.it
- personal website: <https://gcappon.github.io>
- public repos: <https://github.com/gcappon>

What is the course about?

Biomedical wearable technologies...

- **E-health or digital health:** The use of modern Information and Communication Technologies (ICT) to efficiently deliver all kinds of healthcare services
 - Examples: EHR, telemedicine systems
- **Wearables:** electronic devices worn close to and/or on the surface of the skin, which detect, analyze and transmit biomedical data, such as heart rate, activity, blood pressure, temperature, etc.
- **Internet-of-thing (IoT) wearables:** Wearables that connect and exchange data with other devices (wearable or not) and systems over the Internet or other communication networks



What is the course about?

... for healthcare and wellbeing

- Our society is facing some important global health challenges
 - **Lifestyle-related illnesses** driven by sedentary life, consumption of processed and fast foods, alcohol drinking and smoking
 - Need to change the paradigm of healthcare, from a mostly reactive system to a **proactive system**
 - **Population ageing** rises the demand of health services and the healthcare expenditure.
 - Need to **decentralizing the healthcare**
 - Covid-19 pandemic called for urgent delivery of healthcare at home

Biomedical wearable technologies are fundamental to make healthcare scalable and accessible outside the hospital setting and deliver personalized and preventive health services

Theory program

- The theoretical part of the course will cover the following:
 - Wearable sensor technologies
 - Health design thinking
 - Elements of UI/UX Design
 - Evaluation of app usability
 - Fundamentals of network architectures and the HTTP communication protocol
 - The representational state transfer (REST) paradigm
 - Application programming interfaces (APIs)
 - Elements of network security
 - Authentication protocols
 - Regulation for health data management

Lab program

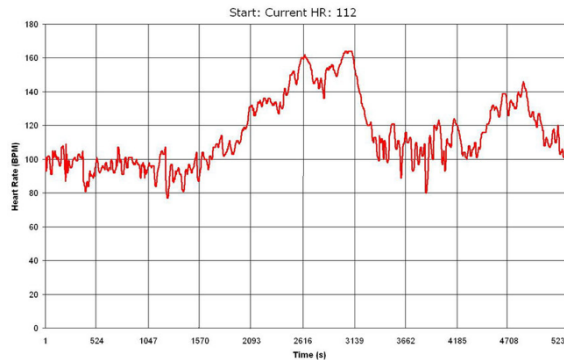
➤ The laboratorial part of the course will cover the following:

- The Git version control system
- Dart programming language and the Flutter framework
- Building mobile/web applications in Flutter
- User Interface design
- Navigation
- State management
- Basic data persistence
- Networking
- Integrating AI models in mobile applications

➤ Workshop...

Lab program - The workshop

- 2-hours workshop where we will set up a fake clinical trial aimed at evaluating the cardiovascular tone in a cohort of healthy subjects using an experimental algorithm



- The fastest 5 students to submit the right answer will get a +1 to the final evaluation.

1. Trial setup

2. Data collection using a wearable and an app

3. Data preparation

Together

4. Data analysis

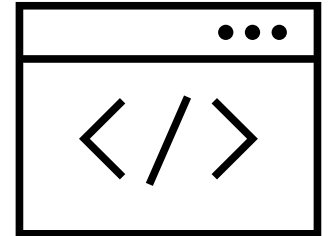
5. Trial outcome

Individually

Prerequisites & End of class goals

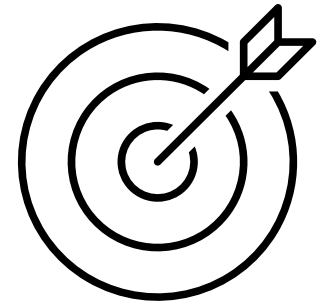
➤ Prerequisites

- Basic:
 - Propension for programming, Python
- Preferable:
 - Object-oriented programming



➤ End of class goals

- Knowing how wearable devices communicates to the end-user
- Designing digital solutions to enhance healthcare and well-being
- Developing a fully functioning mobile application



The exam

➤ **Written test:** Theory exam

- 4 open questions
- Duration: 90 min
- Max vote: 20/30

Dates of the written test

- First attempt: 18 Jun 2026, 14:00 (Room Me+Ne)
- Second round: 07 Jul 2026, 14:00 (Room De)
- Third round: 28 Aug 2026, 14:00 (Room De)
- Fourth round: 01 Feb 2027, 14:00 (Room Ee)

➤ **Project:** Development of an application that integrates and elaborates data from a smartwatch. To be done in groups of 3 students.

- Duration of the discussion: 45 min (25 min presentation + 20 min Q&A)
- Max vote: 12/30

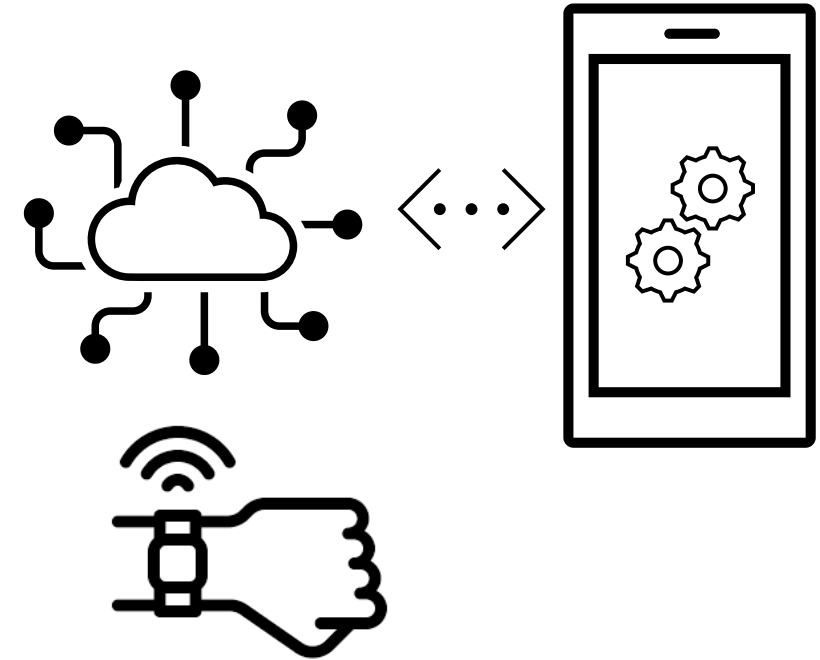
➤ **Final mark:** sum of the votes of the written test and the project.

Project: What's the spirit here

- The goal is learning “**How to do SOMETHING ORIGINAL using wearable devices**”
- What **I DO** expect from you
 - To use your imagination to create that SOMETHING
 - To learn to build something from scratch
 - To use code built by others
 - To work together
 - To use Google and StackOverflow
- What **I DO NOT** expect from you
 - To create the new Google
 - To reinvent the wheel (copying others code is fine)
 - To create extra complicated UI with animations and stuff

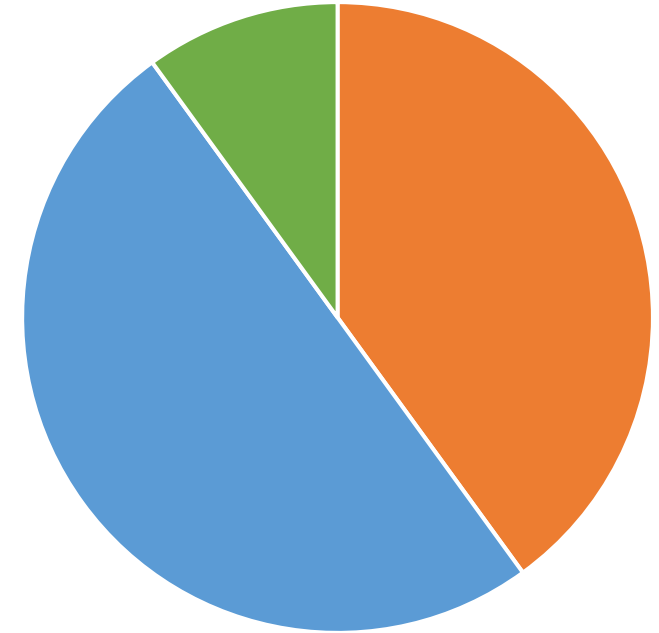
Project structure

- The project consists of building an app for iOS or Android that collects user data from a wearable device through Web APIs, visualizes them, and does some tricks with them.
- Core functionalities:
 - Correct app flow
 - User authentication and management
 - Data collection
 - Data visualization and presentation
- Additional functionalities → It's up to you!
Some examples:
 - Run some analysis on data and provide suggestions to the user
 - Implement some literature algorithm
 - ...



Grading

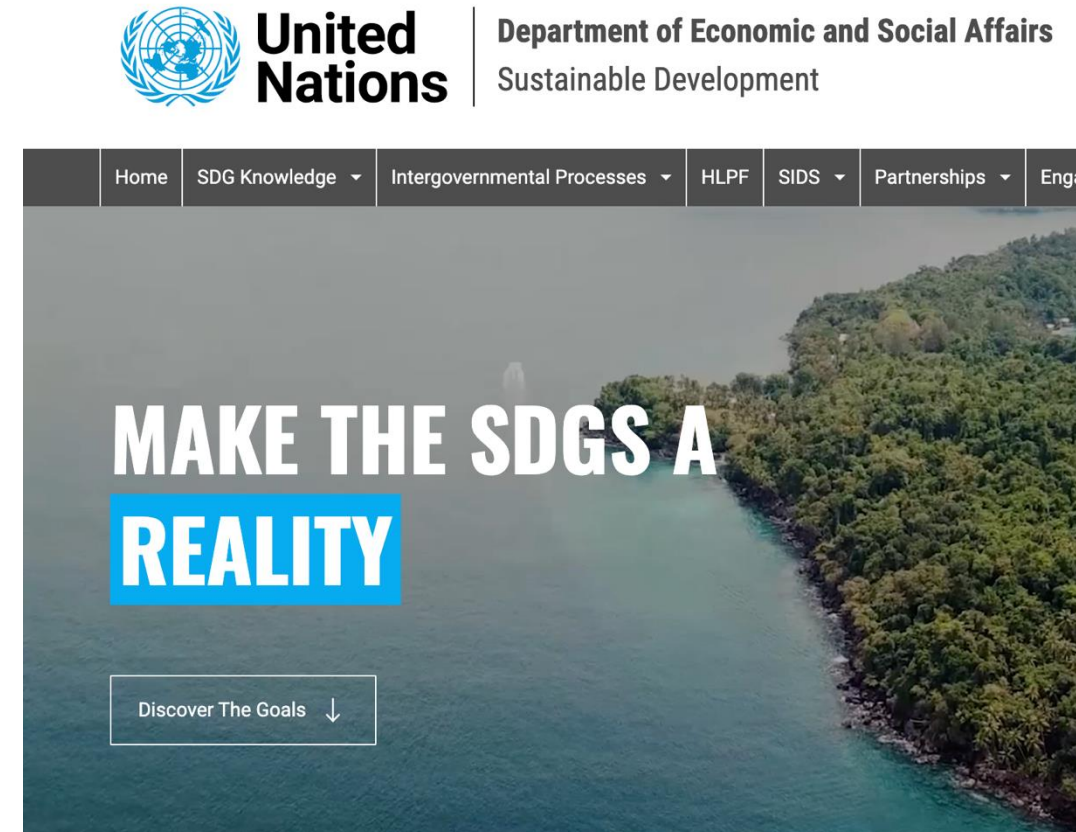
- What are the grading criteria
 - Compliance to core functionalities
 - Originality of the additional feature
 - Quality of teamwork
- How much these criteria matter?
 - Core functionalities: 40%
 - Additional app functionalities: 50%
 - Quality of teamwork: 10%



- Core functionalities
- Additional functionalities
- Quality of teamwork

Choose your target

- Your app must represent a solution for the current SDGs (Sustainable Development Goals) targets
- SDG are 17 topics representing the current big challenges humanity must face and solve



Choose your target

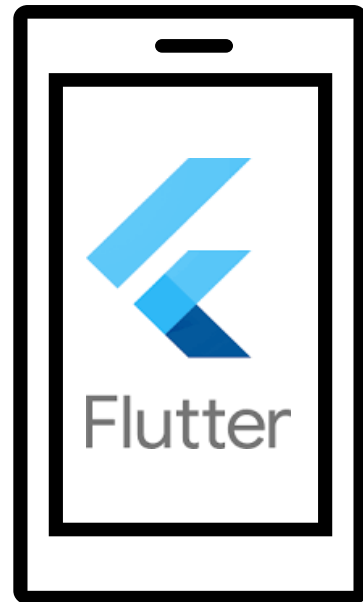
- These are the target you can choose from (see the elearning page for more details):
- **SDG 2: Target 2.2** (By 2030, end all forms of malnutrition, including achieving, by 2025, the internationally agreed targets on stunting and wasting in children under 5 years of age, and address the nutritional needs of adolescent girls, pregnant and lactating women and older persons)
 - **SDG 2: Target 2.3** (By 2030, double the agricultural productivity and incomes of small-scale food producers, in particular women, indigenous peoples, family farmers, pastoralists and fishers, including through secure and equal access to land, other productive resources and inputs, knowledge, financial services, markets and opportunities for value addition and non-farm employment)
 - **SDG 3: Target 3.4** (By 2030, reduce by one third premature mortality from non-communicable diseases through prevention and treatment and promote mental health and well-being)
 - **SDG 3: Target 3.5** (Strengthen the prevention and treatment of substance abuse, including narcotic drug abuse and harmful use of alcohol)
 - **SDG 4: Target 4.7** (By 2030, ensure that all learners acquire the knowledge and skills needed to promote sustainable development, including, among others, through education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship and appreciation of cultural diversity and of culture's contribution to sustainable development)
 - **SDG 8: Target 8.9** (By 2030, devise and implement policies to promote sustainable tourism that creates jobs and promotes local culture and products)

Data source

- Data that you will use are collected by me using a Fitbit Versa 2
- Data are available starting from 09/02 and will be collected up to 30/11
- The following data “types” are included(see mor details in the Appendix):
 - Calories
 - Distance
 - Exercise sessions
 - Heart rate
 - Resting heart rate (Estimated)
 - Sleep
 - Steps
- Your project must use at least 1 type of data



How to get data - The IMPACT backend



Your app

HTTP
requests/responses



<https://impact.dei.unipd.it/bwthw/>

- You will be provided with credentials (different for each group) and learn how to get data during the lab lessons

Optional tutoring activities

- Need help? They are here to cover your back with answers and optional tutoring activities



➤ Luca Cossu

luca.cossu@phd.unipd.it



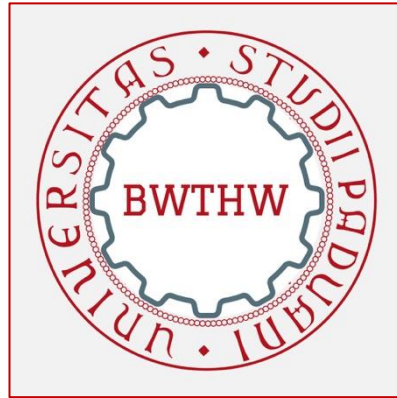
➤ Alberto Gastaldello

alberto.gastaldello.2@phd.unipd.it

Calendar of the optional tutoring activities

Day	Time	Room	Topic
March 7, 2025	8:30 - 10:30	Me	Tutoring 1: Exercises in Dart
March 14, 2025	8:30 - 10:30	Me	Tutoring 2: Exercises in Dart
March 28, 2025	8:30 - 10:30	Me	Tutoring 3: Sample App Presentation + Setup
April 11, 2025	8:30 - 10:30	Me	Tutoring 4: UI + Navigation
April 28, 2025	14:30 - 16:30	DEI/D	Tutoring 5: State Management
May 9, 2025	8:30 - 10:30	Me	Tutoring 6: SharedPreferences + Authentication
May 16, 2025	8:30 - 10:30	Me	Tutoring 7: HTTP Data request
May 23, 2025	8:30 - 10:30	Me	Tutoring 8: Algorithm
June 4, 2025	8:30 - 10:30	Te	Tutoring 9: Wrapping up

Discord server



+



➤ Join the Discord server to get in touch and ask questions

➤ Link: <https://discord.gg/j76A3D27bm>



Appendix: Data source characteristics

Focus on data – Calories

➤ Calories: a list with 1 data point exactly every minute

➤ Fields meaning:

- time: timestamp (hh:mm:ss format) of the calory entry
- value: the calories spent during the last 1 minute (rest + active)

```
[  
  {  
    "time": "00:00:00",  
    "value": "1.29"  
  },  
  {  
    "time": "00:01:00",  
    "value": "1.29"  
  },  
  {  
    "time": "00:02:00",  
    "value": "1.29"  
  },  
  {  
    "time": "00:03:00",  
    "value": "1.55"  
  },  
  {  
    "time": "00:04:00",  
    "value": "1.29"  
  },  
  {  
    "time": "00:05:00",  
    "value": "1.29"  
  },  
  {  
    "time": "00:06:00",  
    "value": "1.29"  
  },  
  ...  
]
```

Focus on data – Distance

- Distance: a list with 1 data point every minute or more depending on user's movement
- Fields meaning:
 - time: timestamp (hh:mm:ss format) of the distance entry
 - value: the distance travelled between the last timestamp and now (in cm)

```
[
  {
    "time": "00:03:00",
    "value": "0"
  },
  {
    "time": "00:50:00",
    "value": "0"
  },
  {
    "time": "01:28:00",
    "value": "0"
  },
  {
    "time": "01:32:00",
    "value": "1440"
  },
  {
    "time": "01:33:00",
    "value": "3110"
  },
  {
    "time": "01:34:00",
    "value": "0"
  },
  {
    "time": "01:53:00",
    "value": "0"
  },
  {
    "time": "03:09:00",
    "value": "0"
  },
  ...
]
```

Focus on data – Steps

- Steps: a list with 1 data point every minute or more depending on user's movement
- Fields meaning:
 - time: timestamp (hh:mm:ss format) of the step entry
 - value: the number of steps done between the last timestamp and now
- **Note:** there can be days without data

```
[
  {
    "time": "00:07:00",
    "value": "0"
  },
  {
    "time": "00:10:00",
    "value": "0"
  },
  {
    "time": "01:57:00",
    "value": "0"
  },
  {
    "time": "02:29:00",
    "value": "0"
  },
  {
    "time": "02:40:00",
    "value": "0"
  },
  {
    "time": "02:57:00",
    "value": "11"
  },
  {
    "time": "02:58:00",
    "value": "18"
  },
  {
    "time": "02:59:00",
    "value": "0"
  },
  ...
]
```

Heart rate

- Heart rate: a list with 1 data point every 5 seconds.
- Fields meaning:
 - time: timestamp (hh:mm:ss format) of the heart rate entry
 - value: the heart rate at the current time (in bpm)
 - confidence: a number from 0 to 3 defining how much the reading is reliable (this probably depends on user movement, etc. You can ignore it)
- **Note:** there can be missing values

```
[  
  {  
    "time": "03:10:19",  
    "value": 58,  
    "confidence": 1  
  },  
  {  
    "time": "03:10:24",  
    "value": 54,  
    "confidence": 3  
  },  
  {  
    "time": "03:10:29",  
    "value": 52,  
    "confidence": 3  
  },  
  {  
    "time": "03:10:44",  
    "value": 52,  
    "confidence": 3  
  },  
  {  
    "time": "03:10:54",  
    "value": 53,  
    "confidence": 3  
  },  
  {  
    "time": "03:11:09",  
    "value": 54,  
    "confidence": 3  
  },  
  {  
    "time": "03:11:19",  
    "value": 53,  
    "confidence": 3  
  },  
  ...  
]
```

Resting heart rate

- Resting heart rate: an entry per day.

```
{  
  "time": "00:00:00",  
  "value": 51.35,  
  "error": 17.86  
}
```

- Fields meaning:

- time: timestamp (hh:mm:ss format) of the resting heart rate entry (always 00:00:00)
- value: the estimated resting heart rate at the current time (in bpm)
- error: the estimate error (in %) (Can be ignored)

- **Note:** there can be days without data

Exercise

- Exercise: a list with 1 entry for each recorded exercise session
- Fields meaning:
 - activityName: the type of entry (“Corsa”: Run, “Bici”: bike, “Camminata”: Walk)
 - averageHeartRate: the average heart rate during the session
 - calories: the active calories spent during the session
 - distance: the distance covered during the session
 - distanceUnit: the unit of the “distance” field
 - duration: the total duration of the session (in ms)
 - activeDuration: the total active duration of the session (in ms)
 - steps: the total number of steps done during the session (if “Corsa” or “Camminata”)
 - logType: how this record was logged (“tracker” or “auto_detected”)
 - heartRateZones: a list that summarizes the number of minutes and calories spent in specific heart rate ranges (from min to max)
 - speed: the average speed (in km/h)
 - vo2max: the estimated running VO2Max (only for “Corsa” activities)
 - elevationGain: the elevation gained during the session
 - time: the starting time of the session (format hh:mm:ss)
- **Note:** different activity types can have (slightly) different fields
- **Note:** there can be days without data

```
[
  {
    "activityName": "Corsa",
    "averageHeartRate": 143,
    "calories": 727,
    "distance": 8.474117,
    "distanceUnit": "Kilometer",
    "duration": 3.548E+6,
    "activeDuration": 3.214E+6,
    "steps": 8604,
    "logType": "tracker",
    "heartRateZones": [
      {
        "name": "Fuori zona",
        "min": 30,
        "max": 110,
        "minutes": 3,
        "caloriesOut": 30.146219999999971
      },
      {
        "name": "Grassi bruciati",
        "min": 110,
        "max": 137,
        "minutes": 8,
        "caloriesOut": 97.13782
      },
      {
        "name": "Attivita aerobica",
        "min": 137,
        "max": 170,
        "minutes": 48,
        "caloriesOut": 659.73843
      },
      {
        "name": "Picco",
        "min": 170,
        "max": 220,
        "minutes": 0,
        "caloriesOut": 0
      }
    ],
    "speed": 9.4981754822650917,
    "vo2Max": {
      "vo2Max": 48.77853
    },
    "elevationGain": 28.042,
    "time": "11:22:43"
  },
  {
    "activityName": "Bici",
    "averageHeartRate": 150,
    "calories": 3893,
    "distance": 140.143695,
    "distanceUnit": "Kilometer",
    "duration": 2.0901E+7,
    "activeDuration": 1.756E+7,
    "logType": "tracker",
    "heartRateZones": [
      {
        "name": "Fuori zona",
        "min": 30,
        "max": 110,
        "minutes": 28,
        "caloriesOut": 214.835279999999842
      },
      {
        "name": "Grassi bruciati",
        "min": 110,
        "max": 136,
        "minutes": 64,
        "caloriesOut": 664.550279999999993
      },
      {
        "name": "Attivita aerobica",
        "min": 136,
        "max": 170,
        "minutes": 232,
        "caloriesOut": 3167.1500100000012
      },
      {
        "name": "Picco",
        "min": 170,
        "max": 220,
        "minutes": 24,
        "caloriesOut": 377.76059999999995
      }
    ],
    "speed": 28.734978587699313,
    "elevationGain": 663.55,
    "time": "11:03:54"
  },
  ...
]
```

Sleep

➤ Sleep: a list with 1 entry for each sleep session

➤ Fields meaning:

- dateOfSleep: day associated to the sleep entry (MM-DD format)
- startTime: the starting timestamp of the sleep (MM-DD hh:mm:ss format)
- endTime: the ending timestamp of the sleep (MM-DD hh:mm:ss format)
- duration: the duration of the sleep session (in ms)
- minutesToFallAsleep: the number of minutes spent to fall asleep
- minutesAsleep: the number of minutes asleep during the sleep entry
- minutesAwake: the number of minutes awake during the sleep entry
- minutesAfterWakeup: the number of minutes the user spent in bed after waking up
- efficiency: the estimated sleep efficiency (from 0 to 100)
- logType: how the entry was logged (always “auto_detected”)
- mainSleep: a boolean indicating if the entry corresponds to the main sleep session
- levels: a summary of the sleeping stages (deep/wake/light/rem/restless) during the sleep session
- data: the stage profile of the sleep session. Each entry contains the starting timestamp, the level and how much it lasted.

➤ **Note:** there can be days without data

```
{
  {
    "dateOfSleep": "02-14",
    "startTime": "02-13 22:44:00",
    "endTime": "02-14 06:36:30",
    "duration": 2.832E+7,
    "minutesToFallAsleep": 0,
    "minutesAsleep": 429,
    "minutesAwake": 43,
    "minutesAfterWakeup": 3,
    "timeInBed": 472,
    "efficiency": 96,
    "logType": "auto_detected",
    "mainSleep": true,
    "levels": {
      "summary": {
        "deep": {
          "count": 5,
          "minutes": 106,
          "thirtyDayAvgMinutes": 86
        },
        "wake": {
          "count": 34,
          "minutes": 43,
          "thirtyDayAvgMinutes": 46
        },
        "light": {
          "count": 28,
          "minutes": 235,
          "thirtyDayAvgMinutes": 281
        },
        "rem": {
          "count": 11,
          "minutes": 88,
          "thirtyDayAvgMinutes": 101
        }
      },
      "data": [
        {
          "dateTime": "02-13 22:44:00",
          "level": "wake",
          "seconds": 30
        },
        {
          "dateTime": "02-13 22:44:30",
          "level": "light",
          "seconds": 150
        },
        {
          "dateTime": "02-13 22:47:00",
          "level": "deep",
          "seconds": 1860
        },
        ...
      ]
    }
  },
  ...
]
```